

Day 15 Hybrid Notes

Choosing a strategy and connecting trigonometry to area | Cloze + outline + active learning

Name:	Date:	Period:
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Learning Targets

I can choose an efficient right-triangle strategy from the given information.
I can use $A = (1/2)ab \sin(C)$ when two sides and the included angle are known.
I can check whether an answer is reasonable and communicate it with units.

Part 1: Cloze Notes - Essential Ideas

- 1. When two sides of a right triangle are known, consider the Pythagorean _____.
- 2. A 45-45-90 or 30-60-90 triangle uses special side _____.
- 3. An acute angle and a side suggest sine, cosine, or _____.
- 4. To find an angle from two sides, use an inverse trig _____.
- 5. For two sides and their included angle, area is $A = (1/2)ab$ _____.
- 6. The included angle lies between the two given _____.
- 7. Context answers should include appropriate _____.

Part 2: Outline Notes - Big Picture

- A. Strategy choice begins with marking known and unknown _____.
- B. Special ratios give exact values, while calculator trig often gives _____.
- C. The trig area formula connects sine to perpendicular _____.
- D. Reasonableness checks catch side-label and calculator-mode _____.

Connection to Prior Math

- 1. What algebra skill shows up today?

- 2. What diagram or graph skill shows up today?

- 3. What is one place this could appear outside math class?

Most Important Reminder

Before solving, label the diagram, name the relationship, and check whether the answer makes sense.

Day 15 Hybrid Notes

Guided setup + active learning

Part 3: Guided Setup Practice

1. A right triangle has angle 52 degrees and hypotenuse 20. Find the adjacent side.

Setup/work: _____

Answer: _____

2. A triangle has sides 10 and 14 with included angle 38 degrees. Find its area.

Setup/work: _____

Answer: _____

Part 4: Error Analysis

A student uses $A = (1/2)ab \sin(C)$, but angle C is not between sides a and b.

What went wrong?

Correct idea:

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Before You Solve Checklist

☐ I labeled the diagram or graph.

☐ I chose the correct relationship or formula.

☐ I substituted values carefully.

☐ I checked whether my answer makes sense.

Part 5A: Draw It

Draw a diagram, graph, or labeled example that matches today's lesson.

Part 5B: Two-Sentence Summary

Sentence 1: Today I learned that

_____.

Sentence 2: One thing I need to remember when solving is

_____.

Part 5C: Application Problem

A triangular garden has sides 18 m and 25 m with included angle 64 degrees. Find its area.

Setup: _____

Answer: _____

Sentence answer: _____